

Marc-Aurèle RIVIÈRE

Data Scientist

📍 Trondheim, Norway

👤 35 years old ✉️ marc.aurele.riviere@gmail.com 🏠 https://ma-riviere.com

🌐 ma-riviere 🌐 ma-riviere 📞 0000-0002-5108-3382



Building software and analyzing data for 7 years, with a focus on statistical modeling, R & Shiny.

Experience

R/Shiny Developer

Remote

Data Champ'

2024 - 2025

Delivered data products end-to-end for clients across multiple industries, from requirements and architecture through deployment and iteration, interfacing with both technical and non-technical stakeholders. Projects included:

- An investment analytics platform for France's development agency (AFD), to monitor their portfolio of billions in international development funding across thousands of projects, with multi-database integration, interactive geospatial visualizations, and automated impact reporting.
- An enterprise-scale energy Measurement & Verification API for the Belgian Government, automating standard and IPMVP methods for energy consumption modeling, statistical analyses, multi-platform data integration, and estimating the impact of renovations on energy consumption.
- A comprehensive HR management system for the AFD to track thousands of employees across multiple branches, featuring recruitment campaign management, and role-based workflows.
- A modular package for a branch of the French government, to integrate multiple sources of heterogeneous geo-spatial data, process them into spatio-temporal indicators at various levels of spatial granularity, and archive them in a specially-designed DB.

Data Science Consultant

Remote

Freelance

2023 - 2024

Sales/Marketing:

- Developed an agentic prospect discovery tool with social media scraping, multi-CRM integration, and lead-database enrichment
- Built a personalized outreach pipeline with RAG over company offerings, and A/B analysis against traditional template-based emails

Bioinformatics:

- Data analysis for 4 published neuroscience papers: hierarchical GLMMs across heterogeneous data types (IHC, imaging, qPCR, behavioral); AR1 structures for time-varying repeated measures; multiple candidate models with PPC & AIC-based selection
- Created app to allow researchers to visualize and analyze their data, with automatic enrichment from NCBI (R/Shiny)
- Built interactive dashboards to showcase results for select publications (R, Quarto, JS)

Research Engineer

Rouen, France

LITIS

2021 - 2023

- Designed & developed an Augmented Reality platform with motion tracking to evaluate and train non-visual navigation (Unity/C#)
- Designed a wearable vibrotactile belt and an interactive haptic tablet for haptic-based navigation for blind people (Java/Arduino)
- Designed psychophysics experiments to evaluate the impact of spatial cues on navigation and analyzed the resulting data (R)
- Developed and tested camera-based solutions for indoor localization (Python/C++)
- Formulated & wrote a 600k€ research grant proposal (ANR project "SAM-Guide", 2021)
- Co-organized international and regional scientific conferences

Graduate Teaching Fellow

Rouen, France

University of Rouen

2017 - 2020

Gave university lectures & practicals on Web Development (JS), GUI design (Java), Image Processing/Computer Vision (Python)

Research Engineer Intern

San Francisco, USA

SKERI

2018

Worked on an indoor localization tool using Visual Inertial Odometry, particle filtering, and object detection (Python, Swift, OpenCV)

Research Engineer Intern

Grenoble, France

LPNC & GIPSA-Lab

2016

Worked on an image-to-sound conversion app for blind image exploration (C++), iteratively improved based on users' performance

Education

PhD - Biomedical Engineering

Rouen, France

Normandy University (unfinished - COVID)

2017 - 2020

- Design, develop and evaluate wearable haptic devices to assist blind people with navigation tasks in augmented reality (Java)
- Design, execute, and analyze multiple psychophysics experiments with human subjects (R)
- Implement algorithms for real-time localization and mapping using embedded/mobile sensors (Python/C++)

MSc. - Computational Neurosciences

Grenoble, France

PHELMA - INPG

2015 - 2016

- Fundamentals of Memory, Perception, Linguistics, Imaging (e.g. fMRI, EEG, MEG), and Robotics
- Experimental design, Statistics, Bayesian modeling, Deep Learning, Signal processing (MATLAB & R)

Skills

Programming

R, Python, SQL, JS, Java, C#, Stan

Frameworks & Tools

Shiny, Plumber, Quarto, Git, CI/CD

DBMS

Postgres, DuckDB, SQLite

DevOps / Infra

Ansible, Terraform, Docker

Languages

French (Native), English (Bilingual), Norwegian (A2/B1), Spanish (A2)

Publications

- Rodriguez-Duboc, A., Basille-Dugay, M., Debonne, A., **Rivière, M.-A.**, Vaudry, D., & Burel, D. (2023). Apnea of prematurity induces short and long-term development-related transcriptional changes in the murine cerebellum. *Current Research in Neurobiology*, 5, 100113.
- Faugloire, E., Lejeune, L., **Rivière, M.-A.**, & Mantel, B. (2022). Spatiotemporal influences on the recognition of two-dimensional vibrotactile patterns on the abdomen. *Journal of Experimental Psychology: Applied*, 28(3), 606–628.
- Coughlan, J.M., Biggs, B., **Rivière, M.-A.**, Shen, H. (2020). An Audio-Based 3D Spatial Guidance AR System for Blind Users. In *Lecture Notes in Computer Science* (Vol. 12376, pp. 475–484). Springer.
- **Rivière, M.-A.**, Gay, S., Romeo, K., Pissaloux, E., Bujacz, M., Skulimowski, P., & Strumillo, P. (2019). NAV-VIR: An audio-tactile virtual environment to assist visually impaired people. *Proceedings of the 9th International IEEE/EMBS Conference on Neural Engineering*, 1038–1041.
- **Rivière, M.-A.**, Gay, S., Pissaloux, E. (2018). TactiBelt: Integrating Spatial Cognition and Mobility Theories into the Design of a Novel Orientation and Mobility Assistive Device for the Blind. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 110–113). Springer.
- Gay, S., **Rivière, M.-A.**, Pissaloux, E. (2018). Towards Haptic Surface Devices with Force Feedback for Visually Impaired People. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 258–266). Springer.

Data analysis contributions

- Rodriguez-Duboc, A., Racine, C., Basille-Dugay, M., Vaudry, D., Gonzalez, B., & Burel, D. (2026). Innovative 3D-image analysis of cerebellar vascularization highlights angiogenic gene dysregulations in a murine model of apnea of prematurity. *The Cerebellum*.
- Rodriguez-Duboc, A., Velicer, S. T., & Kobro-Flatmoen, A. (2025). Bnip3 expression in the brain of an Alzheimer's disease rat model. *Brain Mechanisms*, 148–150, 202518.
- Leroux, S., Rodriguez-Duboc, A., Arabo, A., Basille-Dugay, M., Vaudry, D., & Burel, D. (2022). Intermittent hypoxia in a mouse model of apnea of prematurity leads to a retardation of cerebellar development and long-term functional deficits. *Cell & Bioscience*, 12(1), 148.

Packages

Argent

 ma-riviere/Argent

R Package

A unified interface to build AI Agents using Large Language Models (LLMs) from multiple providers, supporting tool calling (functions, server tools, and MCP servers), multimodal inputs, and structured outputs.